Requirements Gathering

Project: Supply Chain Analytics Dashboard (Power BI)
Program: DEPI, Data Analysis Track
Team size: 6 members
Data source: `supply_chain_data.CSV`, a single flat table of 100 SKUs and 24 columns

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1. Project Context

The project turns a supply chain dataset into an interactive Power BI report that helps decision makers monitor inventory, supplier quality, logistics performance, cost, demand and lead time. The report is organized into four pages:

1. Executive Overview
2. Inventory & Supplier Quality
3. Logistics & Cost
4. Demand, Production & Lead Time

> **Note:** The dataset has no official data dictionary and appears to be synthetic. The stakeholders below are the roles that would typically use such a report, and column meanings are inferred from column names and values.

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2. Stakeholder Analysis

Stakeholder	Role in the Supply Chain	Main Need	Interest	Influence
Executive / Senior Management	Overall decision maker	One-glance view of revenue, quality, inventory and shipping health	High	High
Supply Chain Manager	Owns end-to-end performance	Spot bottlenecks and risks across suppliers, stock and logistics	High	High
Inventory / Warehouse Manager	Manages stock levels	Know which SKUs are at risk of stockout	High	Medium

Stakeholder	Role in the Supply Chain	Main Need	Interest	Influence
Procurement / Supplier Manager	Manages supplier relationships	Compare suppliers by defect rate, inspection results and lead time	High	Medium
Logistics Manager	Manages shipping	Compare carriers, transport modes and routes by cost, speed and revenue	High	Medium
Quality Assurance Manager	Oversees product quality	Track defect rates and inspection outcomes	Medium	Medium
Finance / Cost Analyst	Monitors cost and margin	Understand manufacturing cost, shipping cost and cost-to-revenue	Medium	Medium
Project Team (6 members)	Builds the report	Clear requirements, clean data, a stable data model	High	High
Lecturer / DEPI Evaluators	Assess the project	Complete documentation and a working report	High	High

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3. User Stories

ID	As a...	I want to...	So that...	Page
US-01	Executive	see total revenue, units sold, defect rate, pass rate, stockout risk and shipping cost on one page	I can judge overall health quickly	1
US-02	Executive	see which product type earns the most revenue	I know which category to prioritize	1
US-03	Inventory Manager	see how many SKUs have stock below order quantity	I can prevent stockouts	2
US-04	Inventory Manager	get a list of the at-risk SKUs	I can reorder them first	2
US-05	Procurement Manager	compare defect rate and fail rate by supplier	I can identify weak suppliers	2

ID	As a...	I want to...	So that...	Page
US-06	QA Manager	see the split of Pass, Fail and Pending inspections	I know how much of the catalog is still unchecked	2
US-07	Logistics Manager	compare carriers on cost, speed and revenue	I can choose the best carrier	3
US-08	Logistics Manager	compare transport modes on cost and shipping time	I can balance speed against cost	3
US-09	Logistics Manager	see revenue by route	I can detect dependence on a single route	3
US-10	Finance Analyst	see cost-to-revenue and unit margin	I can find SKUs where manufacturing cost exceeds price	3
US-11	Supply Chain Manager	see demand by location and customer demographics	I can understand where and to whom we sell	4
US-12	Supply Chain Manager	see production utilization and lead times by supplier	I can check whether production covers demand and find slow suppliers	4
US-13	Any user	filter every page by product type, supplier and location	I can focus on the segment I care about	All
US-14	Team member	see data quality flags	I know which numbers need caution	Data prep

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4. Use Cases

ID	Use Case	Actor	Precondition	Main Flow	Outcome
UC-01	View executive summary	Executive	Report is published and the user has access	1. Open the report. 2. Read the KPI cards. 3. Use slicers to narrow the view.	Overall performance understood
UC-02	Identify stockout risks	Inventory Manager	Cleaned data is loaded	1. Open page 2. 2. Read Stockout Risk %. 3. Open the at-risk SKU table.	List of SKUs to reorder

ID	Use Case	Actor	Precondition	Main Flow	Outcome
UC-03	Evaluate suppliers	Procurement Manager	Cleaned data is loaded	1. Open page 2. 2. Compare defect and fail rates by supplier. 3. Filter to one supplier.	Weak suppliers identified
UC-04	Compare logistics options	Logistics Manager	Cleaned data is loaded	1. Open page 3. 2. Compare carriers, transport modes and routes.	Preferred carrier and mode chosen
UC-05	Review cost and margin	Finance Analyst	Cleaned data is loaded	1. Open page 3. 2. Read cost-to-revenue and unit margin. 3. Inspect negative-margin SKUs.	Loss-making SKUs identified
UC-06	Analyze demand and lead time	Supply Chain Manager	Cleaned data is loaded	1. Open page 4. 2. Review demand by location and demographics. 3. Review lead times and utilization.	Demand and lead time bottlenecks understood
UC-07	Refresh the report	Team member	Source Excel file is updated	1. Replace or update the source file. 2. Refresh in Power BI. 3. Verify KPI totals.	Report shows current data

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5. Functional Requirements

Priority uses MoSCoW: **M** = Must, **S** = Should, **C** = Could.

5.1 Data Preparation

ID	Requirement	Priority
FR-01	Load `supply_chain_data.xlsx` into Power BI through Power Query	M
FR-02	Set the correct data type for every column	M

ID	Requirement	Priority
FR-03	Rename `Lead times` and `Lead time` to clear names (for example `Order_Lead_Time` and `Supplier_Lead_Time`) and document the assumed meaning	M
FR-04	Add flag columns for data quality issues (for example inconsistent inspection result, manufacturing cost above price) instead of deleting rows	S
FR-05	Optionally split the flat table into a fact table and dimension tables (star schema)	C

5.2 Report Pages and KPIs

ID	Requirement	Priority
FR-06	The report shall contain four pages: Executive Overview, Inventory & Supplier Quality, Logistics & Cost, Demand, Production & Lead Time	M
FR-07	Executive Overview shall show Total Revenue, Units Sold, Avg Defect Rate, Inspection Pass Rate, Stockout Risk % and Avg Shipping Cost	M
FR-08	Executive Overview shall show Revenue by Product type	M
FR-09	Inventory & Supplier Quality shall show Stockout Risk %, Stock-to-Sales Ratio, Avg Stock Level and a table of at-risk SKUs	M
FR-10	Inventory & Supplier Quality shall show Avg Defect Rate, Fail Rate and Inspection results by supplier	M
FR-11	Logistics & Cost shall show Avg Shipping Time and Avg Shipping Cost by carrier and by transport mode	M
FR-12	Logistics & Cost shall show Revenue by Route	M
FR-13	Logistics & Cost shall show Cost-to-Revenue Ratio and Unit Margin	M
FR-14	Demand, Production & Lead Time shall show Production Utilization, Revenue by Location and Units Sold by Customer demographics	M
FR-15	Demand, Production & Lead Time shall show Avg Supplier Lead Time, Avg Manufacturing Lead Time and End-to-End Lead Time	S

5.3 Interaction and Delivery

ID	Requirement	Priority
FR-16	Slicers for Product type, Supplier and Location shall be available	M
FR-17	Navigation buttons shall link all pages	S
FR-18	Tooltips shall explain KPI definitions	C
FR-19	All DAX measures shall live in a dedicated Measures table with a page prefix (for example `INV_`, `SUP_`, `LOG_`, `COST_`, `EXE_`)	M
FR-20	The final report shall be published to the Power BI Service and the `.pbix` file uploaded to GitHub	M

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6. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Every page shall load and respond to slicer changes within a few seconds (the dataset is small, 100 rows)
NFR-02	Accuracy	The same KPI (for example Total Revenue) shall show the same value on every page
NFR-03	Data integrity	Rows shall not be deleted during cleaning, and known data issues shall be flagged and documented
NFR-04	Usability	Every page shall follow one theme, layout and color palette, and visuals shall have clear titles
NFR-05	Readability	The report shall be designed for 16:9 display with readable font sizes and high contrast
NFR-06	Maintainability	Naming conventions, the Measures table and the cleaning steps shall be documented so any team member can continue the work
NFR-07	Compatibility	The report shall open in Power BI Desktop and display correctly in the Power BI Service

ID	Category	Requirement
NFR-08	Security and privacy	The dataset contains no personal data, and access to the published report shall be limited to intended viewers
NFR-09	Collaboration	Work shall be split by page, with one team member locking the base data before pages are built
NFR-10	Documentation	All project documents shall be stored in the GitHub repository

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7. Assumptions and Constraints

Assumptions

- `Lead times` is treated as order lead time and `Lead time` as supplier lead time. This is not confirmed by a data source.
- `Costs` is treated as an overall cost figure, but it is unclear whether it includes shipping and manufacturing costs.
- End-to-End Lead Time is calculated as Supplier lead time + Manufacturing lead time + Shipping times.

Constraints and data limitations

- The dataset has no date column, so trends, year-over-year analysis and on-time delivery (OTIF) cannot be produced.
- `Revenue generated` does not match `Price x Units sold` (correlation about 0.07), so Revenue-based KPIs need caution.
- 44 SKUs have manufacturing cost higher than price.
- Inspection results are inconsistent in some rows (for example a Pass with a defect rate above 4%), and 41% of SKUs are still Pending.
- 31% of customer demographics values are Unknown.
- The dataset has only 100 rows and weak correlations, so conclusions rely on group comparisons.

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8. Requirements Traceability

Page	User Stories	Functional Requirements
Executive Overview	US-01, US-02, US-13	FR-07, FR-08, FR-16

Page	User Stories	Functional Requirements
Inventory & Supplier Quality	US-03, US-04, US-05, US-06	FR-09, FR-10
Logistics & Cost	US-07, US-08, US-09, US-10	FR-11, FR-12, FR-13
Demand, Production & Lead Time	US-11, US-12	FR-14, FR-15
Data preparation	US-14	FR-01 to FR-05